

Alex Kiefer ~ Active Inference Insights 014 ~ Representations, Predictive Coding, Teleology

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hello everyone and welcome back to active inference insights today I have the great pleasure of speaking to Alex Kiefer Alex is currently the senior research and design engineer at Versus uh but was previously a lecturer at the University of Munich in philosophy his work is deeply exciting and unique and focuses on the convergence of machine learning computational neuroscience and philosophy of mind which Alex I'm sure you know some of my favorite topics the latter one I feel a little bit more well versed on but I can't wait to learn more uh from you about the two prior um firstly thank you so much for joining me this is really exciting I think I think you know reading over your work both prior to the kind of research that I did for this and during I found it to be incredibly as I say unique and special because I feel like you're one of a kind in so far as not many people are really being really able to merge the mathematical side and the philosophical side so I'm actually just curious more generally about how you got into this convergence point was it through maths and uh machine learning or was it more through philosophy and then you had to add the maths on top of that yeah well first of all thanks so much for uh for talking with me like I'm looking forward to just discussing with you and it's nice that it also happens to be on a public uh podcast um absolutely um so so yeah that's a good question uh so I I came from a philosophy actually originally came from an art a fine arts background so I'm I have I have a tendency to go from one thing to another but I got into all of this through philosophy so I would say as a philosophy graduate student I was thinking about cognitive architecture you know in a naive way I was starting to think about these things uh and this was um you know this was this was some time ago so I'd say the landscape was a bit different so not every philosophy graduate student would necessarily have been exposed to like deep learning connectionism um indeed when I was working on my PhD deep learning was just starting to sort of be a thing uh so I started I started to um find references to like connectionist models for things like natural language processing uh and I got interested in those things um and I'd say I mean as is the case for many

people in in my field um Andy Clark's um BBS article whatever next big water moment where I i' had been thinking about um for example um high order basian um basian models uh signal detection Theory Of Consciousness for example by people like hakwan Lao um so I've been thinking in these terms of generative models already uh but then that article really just sort of caused some things to click into place for me um and I'd had sort of a um I don't know amateur or casual interest in um software development coding on the side and Mathematics um so essentially I got into this through philosophy I picked up as much mathematics as I could off on the street as it were uh to do the things I wanted to do and I just thought it was really essential to be able to build some of these models to understand really what the hell was going on right so if uh not not to not to get too far into this topic yet but um I mean part of my philosophical point of view is um is that really a naive sort of uh atomistic symbolic approach to content uh the content of mental representations or representations in general is not really um is not really the way to go although it's it's sort of um maybe predominant I'd say in a lot of the philosophical literature so um to really understand something I think you have to understand its functional profile and so I I thought that there were a lot of discussions in philosophy about cognitive architecture that we're not sufficiently grounded in um a kind of knowledge of the the way that these representations might work which I think to some extent I mean there's always um sort of the low road and the high road right there's always top- down understanding and more bottom-up understanding but I think some bottom up understanding um I felt was necessary to to understand this this stuff and then um of course once you start to work with models like this they um sort of become interesting to work on their in their own right so to some extent I was pulled away from the more purely theoretical questions into wanting to build better models of cognition sure well I mean it's in the most praise uh you know praising way possible that you wouldn't be able to tell hence why I had to ask the question um I mean again I'm not a mathematician so I can't you know I can't say you know I can't be an authority uh on your mathematical chops but honestly it's um I've always I've always sort of considered you a philosopher and then I you know reread the papers I was like wow some of this is like super technical uh I was really really impressed so no I find it very I find it very admirable when philosophers because I guess it's a path that I'm Al you know to be candid also trying to pursue which is supplementing or combining theoretical thinking with um you know with mathematical modeling or mathematical formalisms so it's it's kind of nice to see you and people like Maxwell pave that pave that way so thanks in a weird kind of indirect way um I'm so glad you brought up content and representation because that's actually

exactly where I wanted to start um I kind of thought we would start somewhat broadly so I guess um it's funny you know doing a psychology Masters you probably hear the word representation quite a lot and I did uh but no one ever takes no one ever really goes deeper than that it's like okay there are representations and we all kind of have a hunch about what that means and we think about paintings or photographs and they stand for something however just doing this podcast and doing more reading you kind of see that it's actually an incredibly contentious issue what a representation is do we have representations in the brain mind uh and so on I was I I feel like i' come across Notions of um similarity and resemblance in the representational literature but maybe not as sort of directly explicated by yourself you sort of speak about a form of basian structural representationism I thought this was a really really interesting idea because as you know and as you recognize in philosophy resemblance or similarity and representation are often divorced um people say well you can't get representation out of resemblance something akin to that how do you manage to get representationalism out of sort of structural similarity uh and you know what do your what does your position entail more broadly yeah um so so indeed I think representation is something that it's it's a term that a lot of people across many fields use and you know I think they're justified in using it it's it's a it's a fairly easy language game to pick up in a way to start talking about representations um and I think I think the term has just has just proven its usefulness just in the the fruitfulness of all the various um research programs that have been that that employ this as a central term um so I guess I guess to start with your direct question um I think one one so right so how do you get how do you get past the normal the usual objections to basing um an account of representation a philosophical Theory representation and resemblance um I guess the I guess the the maybe the first objection that you might encounter is well resemblances all over the place um it's very cheap um whereas representation especially in the sense that's relevant to psychological um capacities and theorizing about those capacities should be something more special so I'd say the first move to make is to say that representation is also cheap um that is to argue that so I actually have a I'll probably say this a lot I I say it in any case enough in conversations I have a paper on this I haven't published um sort of against inflationary concepts of um mental representation so I think there are some D you know not dangerous there are some Waters you get into here um flirting with things like pan psychism that maybe we can touch on at some point but um I think I think here here's my view if if you can come up with a serviceable notion of refer presentation that plays the that that does the explanatory work it should both with respect to

cognitive science and also also in common sense terms with respect to like uh capturing the the Manifest image as well as the scientific image right if you can sort of understand how this this thing that you're calling representation does those jobs and if it turns out that it's everywhere well then it's everywhere right so I don't think there's any need to frontload the assumption that representation needs to be rare um so that's that's what now I don't want to say that every resemblance relation is a representation relation but I think you can part of the answer to to that objection is to bite the bullet and say you know individual neurons probably represent things for example uh in in the same sense in the same fundamental sense as you or I do at the personal level maybe with different um features within that right within that category but we still get the extra step right so we still get the extra step from purism resemblance to in your in your work at Le I mean some people would just say there are you know at least in basic Minds there are no representations and here I'm thinking about the sort of radical and activist who say it's only present in these kind of scaffolded non-basic minds or or basic forms of mind uh but you do want the positive representations as you said and and you do make it clear that we have to go beyond resemblance so maybe what I'm kind of leading you here is is towards the kind of Teo semantics or a kind of more functional semantics Tex I really enjoy I I mean I personally really really like this idea um that in some ways a representation means something to the organism I guess because it for me it feeds into ideas of um sense making or affordances or whatever you want to call it so how do we go from Pure objective in quote marks res resemblances to a representation that is for or realized within an organism yeah so all right so I'm going to try to spell this out this is difficult to me this is like in a way it's closest to the core the very core philosophical problems that I think remain sort of unresolved or without a uh uncontentious Revolution resolution for good reason because because it relates to basically the way in which the sort of subject relates to the object the Mind relates to the world so this is the caveat this is still an ongoing um I don't feel settled about about some of these questions um what what what yakob uh Hui and I said in that in in our papers on this um is that um following more or less the account that had been in in the literature um is that you need not just structural resemblance in order to have representation but exploitable structural resemblance um but I do want I do want to emphasize that um I and I can to some degree speak speak for yakob since we collaborated on this but you know he has his own views um so I'll just speak for myself for now um I I don't see either the resemblance relation or the exploitability relation as necessarily um in fact relations to concrete particulars external to the organism uh let's start with resemblance um so and this this is uh this is in the papers that we

wrote although it could stand to be made more explicit um I think um the the fundamentally the kind of resemblance I think that the account rests on is um resemblance between the structure of your internal model um and some hypothetical world and it needn't be the actual World um so what you represent in the first instance and and this this sort of um resonates with some of the things that um were said early on in the basian brain as the spasi and brain ideas started to break across philosophy and cognitive science about um you know your representation of the world being something like an internal simulation or perception is like veritcal hallucination or online controlled hallucination MH um so so the resemblance in the first instance I would say is between the internal cognitive structures and a hypothetical World which may or may not uh match the actual world and then I would say a similar thing about exploitability um exploitability you might try to catch that catch that out in very obviously naturalistic terms in ter in terms of like as you said teleology um what is the function of this cognitive structure what does it tend to allow me to achieve or what's what did it evolve to allow me to aeve achieve for example um I am definitely not a Teo semanticist actually I think um there's been a lot of good efforts to integrate active inference and teos semantics in the literature um I I think we can get into this if you'd like but I think it's a I think it's a misstep I don't think you should have to look uh at least if you interpret teleology in terms of um something historical I think it's a misstep I don't think you need to look to any kind of evolutionary history or beyond the current functioning of the organism in order to say what it's the contents of its states art that was already a mouthful I can always go on but there there's plenty there so I guess the first thing I want to just touch on is so yeah you do speak about the sort of structural resemblance between and and this is actually as you say quite at least in my eyes well established within the literature the the the structural resemblance between the Bas the generative model and the generative process I really like your addition there it doesn't have to be the real world because you've got imagination memory dreams and so on and I have had conversations with people who say that I think you know the perception Action Cycle itself needs to be maybe buttressed by uh either retrospective or prospective ways of thinking and that's actually kind of in the phenomenology as well right hle distinguishing between the live moment and then also retrospection and prospection more formally so that's a really nice addition what what is maybe slightly more contentious is if we fold truth conditions in uh or satisfaction conditions and I I feel like again I don't want to speak for you you can tell me that if the generative model is meant to accurately map onto the generative process in a way that minimizes that KO Divergence there has to be some element of Truth tracking or uh we can't

be completely making it up on the spin because then our models are basically useless over time so how much do you then fold not only uh the the real- time exploitation but also the fact that there's got there's some truth relation between uh world and model yeah so so I think on the fundamental um again uh if you just want to characterize what about this thing makes it a representation I think it does I I tend to think in terms of Truth conditions I actually think truth conditions are a nice fit with a basian um approach because what are distributions over like events right and they you can think of these as events and and their descriptions of states of Affairs not it's not like your distribution is over dog right it's over there as a dog so it's it's a nice fit and of course you can think of um of um well there are nice relationships between logic as usually defined over propositions with truth values and probability Theory so I think that all works out very nicely um so the I guess the question is to what I mean practically speaking your your model and this is sort of a very free energy principle type statement I mean your model your model will not exist if it doesn't track the truth to some degree right and actually I know there's there's there's definitely a push back in the um pre energy principle and active inference literature against the idea that these models have to be veritcal because they're used for control of course as well um but I think I think the the clear truth is that the your you're of course so perfect veridicality um at least at the sort of uh highest possible like um resolution or Fidelity of representation is is not needed and it's probably would be counterproductive right you don't want the map to be as rich as the territory usually right but I also think you can't you absolutely need to to track the truth to some degree and the nice thing about structural resemblance approaches is that you can uh you can quantify various ways of doing it but you can try to quantify the degree of structural resemblance and you don't have to talk about just like truth or falsity this this is something that can come inrees sure okay so that's that we're kind of in agreement there there's there's only a slight it's not a caveat it's a kind of red herring I'm sure that we can skirt it's actually something I raised with Carl in our first conversation they haven't raised since because uh you kind of get you kind of get integrated into the basian game and then you forget all of your sort of actually quite useful naive questions you know I asked him Wham if there is just like no numon right wh if there just are no hidden States if I just take a very kind of George Barkley idealist position there's no kind of all there are ideas I mean I guess you could posit the ideas as being the latent States and and their appearance as being the sensory uh you know input but it's it's a kind of classical philosophical question which is whams if we're a brain in the vat do we still get truth conditions out of that is can we say that you know we've got hierarchies of observation and

what they track even within a kind of illusory space yeah I love that I love that question so okay I mean there's different layers of this there's different ways of thinking about these skeptical scenarios um I I would say that you still have truth conditions for for sure to the extent that I mean to some to some degree I'm just like an intuitive internalist or you know not solipsist right but internalist I start from the internal cartesian kind of point of view and um I do think there's something sort of stoic you might say about the way we represent the world like it seems it seems as though to me uh that I perceive things as being thus and so there's sort of a right and the condition for that being true I I think that that so that representation has a truth condition um whether or not of course whether or not it's true or could be true whether or not the structure of the universe is such as to you know um to be true even possibly um I do think uh with respect to any skeptical there might be versions of skepticism where you say there's literally just nothing out there and um I'm not quite sure what to say about that but I have thought more about um so David Chalmers has this great paper called I think it's called The Matrix as metaphysics uh where he goes into some of these issues and and um I don't know if I'll be able to fully justify this at the moment but I know that where I've landed on this thinking about this stuff in the past is that um from the point of view of trying to model the generative process right so you figure every agent has its own sort of subjective ontology which is part of this it's an aspect of this internal simulation that you develop over time through sensory feedback um now whatever the nature of the the numon is um it's something that um through interacting with it has allowed you to generate these structures uh and I think those structures um those structures I guess this a form of structural realism right so the the struct exist um regardless of whether even if we're in a simulation um yeah there's a sense in which the internal representations that I have there's truth conditions would be satisfied unless right so so it could be part of your overall model that you say and this isn't a computer simulation or or something right so so these very high level um sort of beliefs about the kind of world you live in your metaphysical beliefs might might uh falsify your um your overall model or not um but could be agnostic on those points sure yeah I mean I think Job actually has a really nice contribution to in his um how to entrain your evil demon paper which is that like the demon is giving us pretty reliable uh you know pretty reliable signals whether it's tricking us or not I mean we we you there's kind of beautiful sensory motor Loop whereby we move and what we sense is changed by what we how we move and how we move is changed by what we sense and we seem to have quite good control over our evil demon I guess like as just um a kind of budding philosopher you always run that argument to the ground which is

that maybe there is no you know maybe there is no simulation there just is pure awareness right um or there just you or there just you or well not you but you know your mind and you're a solipsist and in that case there is nothing to track it's not like a computer simulation you mentioned David Chas he's wrar this work on virtual reality which I'm sure you've checked out and you know he doesn't think there's such he doesn't think it really matters right like the the same structures the same philosophical relations still hold whether it's uh carbon or silicon or whatever but there's still this kind of binary just distinction okay you've got what is mapping or the generative Pro the generative model and the generative process that which is being mapped and I just think there is at least philosophically I might be wrong here a skeptical position where you just go well maybe there just is the you know the sense of mapping but there's actually nothing behind that right like all there is is the map there is no territory that I'd say yeah so even the of thin sort of realism you get out of this David Chamas perspective might be sunk if it's like I forget what this Russell called it but basically the the the hypothesis that the world sprang into existence um a second ago right and all your memories with all your memories intact yeah buron Russell right did he believe so Russell yeah yeah yeah so yeah so I don't have an answer to that um I don't think they're I in general though I think um yeah so so skepticism is something that indeed you have you have to kind of struggle with it I guess as an if you study philosophy in that way but I I I think to me the the the outcome of that struggle is a just a realization that that's the way things are and uh and that's okay um it's not really okay right because if you actually found out that one of these scenarios were true and you suddenly woke up and you're a completely different world okay that would be quite traumatic um traumatizing but well so would be being alone in the universe you know that would be quite traumatic as well honestly it's it's yes um in most moods it's easy to dismiss but if you're if you take it seriously honestly that could be one of that's that's one of the most terrifying prospects I think but let's acknowledge it rather than um um trying to dismiss it via a um you know superficial argument sure sure I think in part this is we don't have to go into this but this is in part a result of philosophical education at least in the Anglo uh Saxon World which is they start you basically on skepticism um they start you on daycard and then you read stuff in Act of inference about perception and they're all saying well we kind of take direct realism as you know for granted you're like what like we like at school we said that was rubbish um I just wonder and and then you know you start hearing all the externalist attacks on internalism so I'm kind of curious about your experience openly and it it sounds like it's a taboo and it really ought not to be and I don't think it is but openly dis or

promulgating or proposing a internalist position it seems to me at least that from the outside a fairly rare position to occupy and the kind of hot thing is to be as externalist as possible albe there is some push back of course against radical and activism so I'm just curious about sort of sociologically how you found it yeah no absolutely it's been it's definitely sort of a um unpopular position within within sort of this this this well I'd say in general as you said in general these days um so exter so right so before I ever got into the active inference stuff um I I I sort of was already fighting the tide against I'd say externalism to me it's just it's a perfectly legitimate set of motivations right that that motivate the externalists right I want to understand the role that representation and related Notions play in like explanations in cognitive science and that's a sort of third person sort of operation and uh you know uh and so you want to like on on what basis would I posit contents well I'm trying to explain you know how this mouse manages to navigate in its environment or whatever right um and there's no reason not to use um sort of external externalist um externally available evidence in doing that um so so that's fine that's about content extion but I I think um I don't know I think so so that that movement I I'm not sure exactly where the roots begin they probably go back as far as anything but um there were these moments in I guess the 70s in the 80s people like Tyler Burge were really pushing anti-individualism uh in the philosophy of psychology um and I think this is just one of those cases where the sort of the baby got thrown out with the bath water or things right this is like you know a pendulum and I think the pendulum swung very far in the direction of externalism but I just I just think that leaves out um some very very basic points like the fact that what we're trying to explain I I would think and maybe this is too tied to Consciousness I'm curious what you think about that but trying to explain an organism's point of VI on the world so I think you want to limit yourself to to that and if you start to legis about what the organism's point of view on the world is by um reading in information about what you know is in their environment but you don't know how they're parsing it or how they're processing the information into cognitive internal structures I think you're in danger of um um sort of just biasing your account of what that of what their point of view by your point of view and um that's a sort of naive way of putting it but I think you can make some of that more more precise I well yeah I think I yeah I don't know how much we need to I think we're kind of we we're singing from the same him sheet here I think um I mean I think I'm in quite a privileged position actually as a relative Layman because I have no skin in the game right I just see things as I see it in terms of my intuitions and I think also starting from Consciousness and privileging Consciousness privileging lived experience at least biases you towards some form

of internalism I mean I had this conversation with VZ inz poito in one of the earliest episodes of this um series and yeah I I'm exactly like you I mean I read Evan Thompson and I read verel and I love it I actually think it's really beautiful and I love this idea of the sensory motor Loop and Evan Thompson actually in mind and life talks about how the mind body problem can be become a body body problem it's the lived body versus the living body but even if you read that chapter on Consciousness and Evan I'm sure would say this and I'm sure he does say it in the actual book you don't get you still don't get the phenomenal State you don't get the phenomenal reality right you still have that distance and this same thing I say with VZ which is that well I'm very happy to say that um affordances are socially structured by our language by our culture and so on but like there's still something it is like for something to afford itself to me and I just I have a real hard time saying that way of seeing the world is is relative it's in some Vicken steinan way the qualia is a kind of oldfashioned idea and let's throw it it just doesn't do the job for me um I've had a bit of a I had a very small chat with yakob about this and I think we're both on the same page which is that no well actually and to be I just don't think anyone's going to be a really true strong Illusionist about the hard problem and and about Consciousness now you can get into your key Frankish and Dan dennet and say well it's about qualia and postulating these private States and again I'm not really well versed enough to understand that but I think you're Absol Ely right starting from philosophy of Mind analytic philosophy of mind and the tradition makes one very very skeptical at least of reductive um a reductive account which I guess is actually also you know sorry for rambling it's it's also kind of weird because then you see some very strong externalists also push back against uh reductivism as I was talking to Julian kin and Julian is very unhappy with the idea that neural states are you and Evan says this as well but they think well actually you can just expand that notion into your sensory motor capacities and your affordances with the environment like no because you're still missing something like you're missing phenomenal reality so I'm kind of curious about what you think about that like where where is it do we get out of the problem by saying you know neurons aren't new but actually if you talk about your wider sociocultural uh you know context and your history we get you or or do we need to fold in phenomenality yeah so I think there are a couple of issues here um so so one strand of this leads right into some other work I've done that the psychophysical identity stuff where so I think so the the sort of um representationalist versus an activist axis is one thing and certainly I fall on the representationalist end of that and I have to admit I probably need to give an activists they do and and try to really really immerse myself in the core literature there but but to me it's just

it's a non-starter I don't see how you can explain just perceptual errors and things like that without appealing to something that says that the world is this way even though it's not and and to me that's already sufficient to be a representation and and I don't encounter many super radical and activists who think that we don't need representations so once you have representations in play then that's that's all I've ever I've ever assumed um so I don't worry about too much about that but then you're also talking about sort of the I guess the issue of whether the phenomenal can be reduced to the physical um things like that and so in in the psychophysical identity paper I and I'm not sure by the way of course I don't as a basian and a philosopher I don't have settled views quite generally yeah but the view that I explore in that paper and that I do still want to pursue as a serious possibility is that you can have something like like a mind brain identity thesis so to me that whatever ever it is it has to do justice to phenomenology yeah um so if we even if it's that you're just your neurons which I I don't think that's quite the story I think probably the scientific account coming from the the life sciences is and um and physics and so on is not is not finished yet and I think wherever we end up with that I don't I think it'll it'll make it sound less absurd whatever it is it's not going to be that we're a collection of neurons necessarily it's probably some some form of self-organization um that that has a sort of natural we can get into that but but essentially um you'll have that will even if that's true we I I think a satisfying account has to be able to explain um in completely compelling terms um why it how how that could be true right so how could my current experience of the world while looking at my screen right now uh talking to you how how the hell could that be the same thing as a bunch of you know neuronal activity and so on and I don't personally think that applying to my wider ecological niche or anything like that is going to help elucidate that I think it's it's more like we need to elucidate the connection between the physical mechanisms the physical mechanism is actually um is actually stacking the deck too much the physical processes and my conscious experience and I think there's some hope that we can do that but it it cuts along very different lines from the inactivist kind of solution yeah well again I I just personally think that that concession is made at least within a lot of in activism even sort of um like alvino's sensory motor theory talks about how sort of neural signatures will tell you maybe why you're seeing red rather than blue or why you're hearing rather than seeing but it doesn't tell you why you're hearing doesn't tell you why you're saying red yeah I I like your optimism um I don't know if I share it if about the about the relationship between the physical and the phenomenal uh but I have a soft I have a sort of um soft appreciation let's say for more esoteric theories of Consciousness whether that be a kind of

conscious realism Allah Donald Hoffman or maybe even pan psychism but again I've got no commitments um we we'll loop back but I wanted to just quickly touch upon this notion of Tio semantics um which was new to me until a couple of months ago and then I go into like was reading about Tios semantics and Tio semiotics and the difference between those but if we just stick with Tios semantics the idea being that sort of from what Telos purpose biological function let's say you can get representation you can get some meaning and you wanted to push back on that in a way because of the historicity so the content um itself is not necessarily determined by your background it's interesting because I kind kind of see I I kind of see it differently although I don't necessarily I don't necess have a good way of articulating it only because we take as an assumption that the prior preferences we have in our active inference models are at least constituted in part by evolutionary priors and I presume that my representation and also coming from the affordance literature you know my representation of a table is very different from an ant representation of a table because the affordance is the differ so what do you mean in terms of siphoning off The evolutionary history from some form of Tios semantics yeah oh great so I mean first I should say I think Tio semantics is an amazing research program like I've read a lot of Ruth milikins work which is foundational to it and it's fantastic work absolutely foundational to where we are today um but I think I mean the big picture is that I just think as okay I'll answer the question you asked which is what do I mean by siphoning off The evolutionary history but just quickly to motiv you have bite some Bulls so I'm to bite the representation everywhere bullet but one one thing that milin explicitly says is so if you have you know there's these thought experiments like I think it's called Swamp Man or something where some some creature appears suddenly out of out of some Quantum fluctuation or something and is identical to a uh physically identical to a human being who would have evolved over millions of years Etc right to have the same structure uh um what miligan says is that basically the the internal structures of that creature um such as the heart brain whatever um shouldn't really even be called that because those right those terms refer to things that have a certain function and that function comes from the selection history so I guess the idea is well this thing just appeared you don't really know what its parts are for you don't really know what any for and I just completely reject that intuition I think it's clear that if you you know you might have to spend a couple minutes analyzing it but if you or talking to to it depend as the case may be but you know you'll you'll see you'll be able to figure out what these things are for it's like well its heart the purpose of its heart is to pump blood in the sense that it's doing that now and it's keeping it alive and I don't think you need to look further back

um but and and so I guess I guess that does start to get into what I mean by by separating these things off so I think if you have if you have a rich enough account of the current functional organization of something that you should be able to figure out there's a sense in which you can cash out uh the the purpose of the structures within that thing without knowing about the history so maybe I'm relying here on an overly um complacent distinction between like constitutive and causal um relationships so like certainly the history is necessary to understand how this structure got to exist I just think that's a different question from like what is it doing now possibly what its purpose now certainly what's its content so I want to add one more remark here which is even if you had um a nice way of Cashing Out biological function um let's say you could even do it in a way that doesn't appeal the history I still don't think that that's even the same thing I don't think that's the what you want to explain semantic normativity I just think it's a different kind of question yes yes and in fact that's interestingly the very point that hotto and my make in rejecting Tio semantics which is that like you how you gain content out of biological necessity and I think Jerry Fodor had a similar stance um yeah I also wonder whether we need to make the distinction between what I can say about the swamp creature and what it is like to be the swamp creature in terms of maybe it's not necessarily for me to know the entire evolutionary past of the swamp creature in some deep epistemological ways to account or or to make a rough approximation of what's going on but as you say in terms of the actual ontological status of its internal States and by that I mean I I I buy the thing about heart and lungs because you can get those by analogy but I guess the idea of the kind of private mental States you probably do get some way by uh accounting for its evolutionary past right in the same way that I get some way from distinguishing myself from an ant because well we are very different parts we've and that evolution is some way I'm not saying entirely but in some way is guiding that which you know the affordances which the landscape presents to us but no I so so if we then go back to actually what you do say your s positive account and give a quote here um I thought this was really great so um so you've got sort of yeah your your structural resemblance and how we're going to get representation out of that and you say that um this problem can be solved by requiring that the structural resemblance can be used or exploited by a cognitive system in order to make cognitive functioning effective um this captures the intuitive idea that a representation mental or otherwise serves as a proxy or standing for what is res uh what is represented and I certainly see that in the sense of how you would then get representations because in a sense it's its meaning for the organism some evolutionary biologist might go well that just you know what

makes something is effective is its evolution you know it's evolutionary payoffs I'm still struggling at least from that to see the strong divorce from Tio functionalism or or Tio semantics is it the addition of the kind of Truth conditions that get us away from a purely Tio semantic or fun or functional semantic semantics well okay so I guess I don't see a deep distinction between an account in terms of Truth conditions and a Tios semantic account necessarily okay I thought at least um that what people like milikin were trying to do is provide an account of how you get from um very naturalistically um understandable things like States carrying information about other states to uh and things like natural selection to an account of representational content and the main ingredient that's missing there is the possibility of error um and I think of those is sort of one and the same right so that the truth condition of a representation is what would have to be the case for it to be accurate or true and uh you know so if you're if you've provided an account of uh error then in a sense you've provided an account of something that can function as a truth condition but I think I I guess I just don't see so I think you do need some account some some account of the possibility of of error um so we did discuss uh some ways that you might get error out in that paper that you're quoting from I think um so you could have you could have an accurate generative model uh that is it's its structure um resembles the structure of the generative process in relevant ways um but it could be misregistered so you could you know you could infer that you're in some state that you're not because of either because of misapplication of the model or just because hey it's a probabilistic model and yeah sometimes what's probable isn't what's actual right right so so there's that's that's one way of of explaining it um in general I think you can also just look at the broader even just it's a bit of a cartoon to say completely synchronously because any process is somewhat temporally extended so I would say history matters only to the extent that you're trying to give an account of a process that might exist in some time scale and so from the point of view of a smaller time scale that might be a historical matter but I don't think you need you should for example you shouldn't have to go deep or at all into evolutionary history I think to give an account of my current perceptual ability um that is to give an account of the fact that I can perceive things and that I have mental states with contents you do have to appeal to Evolution to explain or or it's it's at least a very good theory if you want to explain like well did I get into the the state in which I have this capacity but I just think those are two different questions sure so I don't think I've really I've really answered your question yet but so I think if you just look at the roughly synchronous um you know present day organization of the organism you should be able to tell what's now this is is this is

sort of idealizing I don't think you can actually do this practically speaking in every case right um You you need to have like a if you could take a snapshot of my physical organization right now and it might take months or years analyzing it to figure out what the actual functional profile of all the components of me as a physical system is or are um if you could do that I think you could tell a story about like well what's the what what is the what is the current functional role of this thing say it's a it's a belief that you know uh belief that there's a fence across the street um and you give an account of the content in those terms in terms of like well which aspect of the generative process does this aspect of the generative model map on to um and yet it could still be the case you could independently investigate whether there is in fact offence across the street and you could say well this representation has that content but he's wrong about that aspect of the world right yeah yeah yeah yeah no you're absolutely right of course you you you certainly make a point of misrepresentation this is um for those wondering this is representation in the prediction error minimum ization framework by yourself and yob um yeah I think another point you make there which is important is that pure functional semantics don't require this structural mapping I think that's important as well that you don't necessarily need this um relationship between something like a generative process and a generative model in traditional Tios semantics um another thing you add that I thought was really interesting I'd love to learn about and sort of peruse is this idea of content holism um because that's a kind of additional part to your proposal and you say here the content of one part of the overall representation depends on the structure of the whole and thus cannot be determined without simultaneously fixing the content of the other parts and you also I should point out in a footnote I believe allow for that to become recursive so that's not you know the the part is not just the part it might also be the whole for subcomponents how what what is Con holism how did you come to that idea and what does it add to your uh structural representationism right so I recently was talking uh with some philosophers uh who are fellow graduates from my um philosophy PhD program at The commity Graduate Center and I um I tend to just like assume that everyone thinks in terms of um you know content holism and stuff and they were like no if you're gonna submit a paper on this stuff you really have to explain that because like people don't so I realized okay that's something has to be unack so um this is something that I got I'd say um primarily from from Quin from the philosopher WV Quin who was um so my my PhD supervisor David Rosenthal was heavily influenced by Quin and sellers and they're both uh both theorists who think along these lines I think this is one of the great divides in like philosophical discussions about semantics um you know in the past

past Century I would just say that um I don't see holism as something that I'm adding to the package it's not like something that I'm sort of tacking on to the view I think pretty much any functional role type of semantics which I take structural resemblance theories to be a species of um is has has to have this feature so um if your if the if the if the view is that your generative model gets its content in virtue of resembling the generative process um and by the way when I just to make this clear because this this also has come up in some discussions with people like um at least when I when I think about that uh I think of the generative model as something that has Dynamics and itself unfolds over time right so so you have a sort of four-dimensional if you will um thing that resembles a four-dimensional external thing or right that might happen to be externally real as well if you're lucky if your model is accurate and and if we're not solists and if yes and if anything exists sorry yeah go ahead yeah so um um right so the um so the structural right so so if you want to understand how anything in that model so so obviously generative models are and generative processes are complex things um if you want to understand the content of any particular mental state say and let's say it's realized by some pattern of activity in prefrontal cortex or whatever you need to understand and how what role does that play in My overall model this four-dimensional object and then what analogous role does something play in the system I'm representing and I just to me that seems like from the ground up it's going to be a holist account because the content of this structure really depends on the role it plays in the whole right right I guess uh where I'm where I probably need clarification is the idea that part part relations as you say themselves are not resemblance relations and then folding that into the footnote which is that the parts can also function as structural representations in these hierarchically organized systems I guess I don't so if the content is fixed according to its representation the structural representation as a whole and thus it doesn't get its content just purely by the part that it plays but actually well the you know the individual part that it plays actually the fact that it's integrated in a hole and yet it can also constitute a hole within let's say a smaller Mark of blanket if you will um those can we hold both of those up at the same in the at the same time um I think so I'm glad you're talking about the footnotes by the way I love my footnotes I would have footnotes within footnotes if I could footnotes are very important footnote people people ignore them I love footnotes yeah they're ter they're terrible for readers but anyway they're useful for Artis yeah yeah yeah so I guess I don't see any any um conflict here so um for examp say some some cortical mini column has a plays a particular role with respect to your to one's entire brain where it clearly represents I don't know the the the location or the pose of some object and my I say represents

that I mean the role that this mini column plays in the overall system is analogous to the role that object plays in the relationships among things in the in the represented system right so that's the basic resemblance um then that that mini column might also have its internal structure um in virtue of which it represents something but I think we're talking about two different levels of description possibly two different in a in a multiscale active inference picture those might be two different agents yep yep and whether I really want to attribute agency to Min columns and cortical Min columns I don't know but like sure the point is it would be this this subsystem that would have its own um structural representationally grounded content would be a distinct thing that would just be a part of this larger system which is me and only in so far as that thing played the role it does in my generative model would it have content would it constitute part of my point of view on the world yes so okay so maybe can articulate a bit better I guess my question here is well one is a kind of more pragmatic question is about like sort of where do we cap off the structured representation as a whole like that you know where do I end in some sense but we can leave we can park that for one second I guess my point is you say structured representation as a whole gets its content via resemblance the same need not be true of its parts these get their content via the overall structural resemblance but the part part relations themselves are not resemblance relations so the let's say the content of the the component part gets it you know gets it gets that content because of the overall structural resemblance but my problem here is okay I of course I can grant you that and that works for me but the problem would be then embedding that in a hierarchy where you say that that part itself constitutes a holistic structure who because you say um we do not mean to rule out that parts may also be structured and function themselves as representations right yeah so but then that has its own holistic identity yeah because it's it by you know by which the its component parts form their identity according to this model so I guess I'm just I can imagine there is a way out but I'm guess I'm slightly confused about how something can't have its content fixed unless it is part of a structured hole and yet is providing a kind of structured hole for subparts unless those subparts in some ways are also contingent on the overall structured Hall and we just have a hierarchy of identity or meaning so yeah so so my current understanding of this I think it would be great if it would be really cool if you could cash out the content so let's say I have a generative model whatever phally stitutes it some internal structures I would say um if we could if we could sort of show what the internal relationships are between that model and lower scale models of individual neurons for example such that the contents let's let's say experience look I'm not attributing Consciousness

to neurons necessarily I think we could we could leave you can divorce Consciousness from representational content to some extent even if you're an internalist right I'm I still think you could talk about U maybe this is part of my deep um the way idiosyncrasies of the way I was trained in philosophy but I'm comfortable thinking about you know unconscious thoughts at the personal level and such so I don't think I think you can talk about the content of uh in of my States divorcing it somewhat from Consciousness so we we it would be really cool if like the content of my current thought had some intrinsic relationship to the content of the thoughts of the neurons right but I don't see that that has to be the case and that's my my answer to you is just um whatever lower level structure it is and so yeah again this structure could play a role in my the overall structure let's just say it's my central nervous system that's the relevant vehicle I don't know that's but say it's that it would be in virtue of the this thing would be basically an atom in a sense with respect to that level of description right all that would matter would be and this this is definitely consistent with a hierarchical active inference point of view what would matter would be its Markoff blanket right its internal States can't matter functionally to the system at that level right because any internal states that produce the same input output profile the same interface would would do just as well um nevertheless that thing might be a whole whose Parts um map onto the parts of some external things right it's just that would be a different system that would be like a system at a lower level and so it the cont I I guess one one upshot of holism it's a bit of a of a hermeneutic type of picture where the content um is sort of relative to the or it's It's associated with the system that you're describing and so the fact that this thing is a holistic vehicle for Content um isn't inconsistent with its being just a part at the higher level because those are two different um sort of content descriptions right content sub Alex and content sub uh cortical column or whatever you seem very unsatisfied with that no I'm not I'm not I'm not I'm not unsatisfied it's it's just a broader point it's kind of difficult to do these things in real time in conversation you know you need to sit down and draw something or have some time think about it we can park that we can come back to it we can we can park it it's uh it's a fascinating idea and I think it's also really important for active inference generally um and predictive code because you can have this kind of very coar grained idea of a hierarchy whereby the top Parts contextualized constraint determine the lower parts but that kind of once you get into the granular level and the philosophically rather vague point which is you know priority you know significance hierarchy priority is a spectrum and at what you know at what point we can cast these strict divisions then I think these questions start to matter which is about these parle relations um

which actually leads me very nicely into predictive coding um so yeah predictive coding in some ways comes along with active inference and the Basi and brain hypothesis I think people have started to be a bit more strict about distinguishing sort of continuous State space discrete State space and P schema and predictive processing but predictive coding is still a very important part of literature I really really your paper where I got a complete history lesson in predictive coding Helm Holtz machines uh Jeff Hinton I just I thought it was great um and people hear these words and terms thr around like Helm Holt's machine generative model discriminative model recognition model and they they probably don't know what it means uh and in a sense they they're relying on an authority like you to hold the meaning um so I thought we could kind of unpick that not only for the audience but also for myself um because I take some of these things just as historical fact so perhaps we can start with the distinction between a generative model a discriminative model and a recognition model sure um so with the caveat that I am also this is also for me because refresh own memory is always a good thing sure um so I'd say let's start with generative versus discriminative discriminative model um so this is within I'm already assuming that we're talking about probabilistic models here um so a generative model is just basically a joint probability density over some variables of Interest right um a discriminative model uh if you were going to try to catch that out as like a um in probab in terms of probability distributions would be a conditional distribution right so if you have some input X um and some label y um this model would be a model of the probability of Y given X right and so you can use this to um you know these things are typically used in I guess discriminative types of tasks like labeling images or something uh classification um generative models are just much nicer and more powerful in many ways I think there's all sorts of I I don't need to rehearse all the all the arguments in favor of them but no don't you don't need so in a from a generative perspective instead of so a discriminative model of um like used for image classification might give you yeah the probability of the label cat given this input image whereas you could instead set up a generative model that jointly models um what what's what's the joint probability of labels X and images Y and it could be based on some conditional um some latent variable um and then you would just invert that model given you can and it's a nice thing about paradigms like this and predictive coding architectures are very good at this you can you could Supply an image and invert the model and get and get predict a label or you could Supply a label and predict an image so on and this and this takes us into the recognition model um people might be familiar with the idea of a recognition density but maybe not a recognition model as the inversion of a

generative model yeah so this is something where I need to um if I'm too if I play too loose and fast with this distinction then people like Carl and Maxwell will take me to task for it so this is one this is one thing that that has come more clearly in focus for me over the years which is that um so the hel Holz machine this this was my my introduction to all this stuff sort of computationally was was just coding up one of those and trying it out and um and and learning about its properties um and so in that model you have an an amortized like a a learned sort of Frozen recognition model which is basically it's your it's your it's your um trained model for inverting the generative model and so by inverting the generative model of course I just mean given some input uh you want to infer some latent state that that represents the hidden state that would have caused that input yeah um so uh you know obviously doing that in general is intractable using na inference so you need you want some you want to use for example variational inference uh great so a q distribution which is your approximate posterior and the recognition model uh in the hel holes machine is just a trained amortized um par parameterization of the recognition density so like you give it some input you have a set of bottomup Weights that are can be that can be distinct from the generative weights that just map into the Laten space so that's your recognition model because it's used for recognition yes and yeah and so from a broader predictive coding cognitive scientific perspective you might want to say well clearly organisms like us Implement some form of recognition model let's on the basian brain hypothesis right let's say we have generative models that go from like things like Concepts to arrays of of simulated sensory data you need some way of getting from the actual data you receive to the concept back to the concepts um and so you can think of that as in a in a broad sense of recognition model the reason I said I have to be careful here is that you don't need a amortized mapping in order to do this in predictive coding architectures you can just run the thing um and do iterative inference yeah just try to minimize the free energy over all the nodes in the netork work given an input and so you have a recognition process without necessarily having um a pre-trained recognition model right because the home HS machine has its learning happens in this kind of uh so-call wake sleep cycle right um which I think some people have linked to human sleep but I I W go um I guess I guess a point of confusion for me now I'm not a computer scientist a coder Etc so you are going to have to be patient with me um is the generative model is talking about these probability densities let's say just over two variables um now that seems to me that it doesn't necessarily prioritize one variable over the other like a discriminative model would which sounds to me that then inverting it doesn't really make sense because you haven't got a starting version so is the generative model when

we talk about a generative model in pred uh predictive coding or active inference the probability let's say the probability of a state or the probability of an observation given a state let's say that sounds more like that very you know the line and bases formula sounds like given your explanation like a discriminative model yeah so thanks for the opportunity for clar to clarify this so um yeah so in a discriminative model I my Paradigm for this is like some feed forward neural network where you've got image layers input and N layers processing and like label layer as output there to try to train a a Holts machine for example to do that instead I would have image and label sort of concatenated or you could think of them as two distinct modalities whatever you'd have these on the bottom layer and you'd build whatever however many layers you want on top of that um and so when I talk about inverting a generative model so I kind of implicitly assume that the generative model uh has um latent variables in it and so the latent variable it's often called like Z or H or whatever right so you supply your input you do some inference onto this latent variable right and that's that's your that's your right internal represent that's your that's your internal States in an active inference idiom that would represent the external States the hidden external States and so so both the things that so instead of a mapping from the one let's think of labels and images as two observation modalities right instead of instead of constructing a mapping directly from one observation modality to another in this kind of model you would have a mapping jointly from the observation modalities to this hidden latent variable and inverting the generative model is inferring the latent variable given the observation I see so it's kind of embedded in a way um it's kind of you're you're get you're discriminating something but within that that itself is overarching a the generative relation does that make sense yeah yes exactly so you could do discrimination right um You can always get these conditional distributions from The Joint distribution so this is just a this is a way of that's that's what's great about generative models they do more I mean of course also harder to train they need you know it's more information that you need to model but if you have one then you can do discrimination um by clamping a label say inferring a latent State and then generating the other modality cool excellent um yes okay perfect so the the hel Holts machine is what 1995 1996 and then friston's predictive coding I think was introduced first in 2005 but there were probably hints of it earlier than that um what are the differences between the two um actually I'm not going to pit Mr Diane over Mr friston so I'm GNA leave it I'm I'm not going to try and ask for a comparison let's just ask for the the differences between the two because people probably hear them aligned but don't know how they're distinguished yeah yeah I mean I think a lot of the stuff I I think Carl and others were working on these ideas and

I think a lot 90s were just there's amazing stuff going on but anyway so yeah so one one difference I already mentioned is that um in the hel Holtz machine you explicitly are training a bottomup recognition model and as you said this is usually well originally first proposal for this was wake sleep algorithm where basically you would you you'd present an input propagate the activities given the current weights and then train the generative weights to be more likely to generate that then you would generate a fantasy top down and retrain the Train the recognition weights to to produce that when you get an input um and it was a way of bootstrapping um two models at once so the the generative model and the recognition model um later on and I I don't know as much about this but um in discussing this with colleagues um it's come to light that um something that worked better was to just was to just always um just always use the bottom up activities to train both models so this is called the re reweighted wake sleep algorithm so that's that's a tangent don't need to get into that but um the wak sleep the hols machine didn't work all that amazingly it was just an it was an amazing proof concept I always thought of it as like a minimal mind because it all the basic ingredients you need sort of imagination perception and so on maybe not action but we can do that as um but so so that's that's one big difference whereas in a predictive coding Network um you've got some Vector of um current hidden State activities um you get an input uh you make a prediction or you you send you send a signal up um to the to the hidden layer um this is the prediction error um and the prediction error is computed with respect to the current activity and you you update the activities so as to minimize the the energy term right which just which is just additive across all the nodes um so you don't there's no recognition model um you can just use the uh you can just use the same the generative weights um you just just use the inverse of those to to map to the hidden State um and so you're not training two distinct models you just train if you're training a model like this you would just train the um you would just train the generative model by minimizing prediction error um other differences um so in the hel Holtz machine there isn't really any there's a certain um abstraction at work in that architecture which is that although so I always intuitively thought of this as as um modeling the fact that you know we know from from Neuroscience that the same neural Hardware is used for perception and Imagination right at least largely overlap um and so the hel machine sort of models that because you use the same units the same artificial neurons to do the bottom up sweep as well as the top down sweep um but those two information flows never interact right you've got a wake phase and a sleep phase and when you're running the model in generative mode you only use top down information and so the authors um Hinton

and Di and all pointed out that this is not really ideal as a model of perception because we know that there are top down effects during perception yep so you don't really want just a frozen amortized mapping um that propagates forward you want to use the topdown information during perception and so predictive coding has that feature so if you could just clamp a bottom level feature Vector whatever whatever the input is and the the um the predictive coding algorithm uh will be integrating information from the priors like sort of online in order to perform the bottom up inference in effect right so you've got you've always got predictions coming down from the top and prediction error coming up from the bottom and these two sources of information are integrated online during perception so it's a much better model in that respect Maybe maybe we can also feed in this idea of um horizontality in predictive coding um I actually came to this through Andy Clark's paper as well whatever next because he makes quite you know makes quite an explicit point that it's not just verticality it's horizontality and almost the horizontality if I if I'm not wrong is really kind of where the inference is happening right it's it's it's kind of the selection of the best hypothesis so maybe you could do a better job than me uh and just explain you know what what does horizontal horizontality introduce to this picture so I don't know if I am sure how uh Andre Clark used that word I assume you mean like lateral connections ofal no no that's it's fun uh so right so definitely there's um there's well so whether or not you have lateral Connections in your model right there's there's other ways of achieving that you can have sort of diagonal influences like you you know you use the representations of this layer propagate information upward and then there's there's a sort of recurrent feedback loop where this layer gets modulated again but you in any case you get this sort of I don't know if it's always win or take all but sort of win or take all Behavior where one of the hypotheses at this level selected you can definitely in some of the earliest predictive coder predictive coding models were literally um yeah were not even hierarchical models they were just like um to to model um I guess efficient information processing in retinal ganglia cells I think where you'd have like the receptive so so you'd have a neuron here and it's in it's its output its spiking activity would be modulated sort of dampened by what was going on in its vicinity right yeah yeah yeah you yeah you made that point I think in the paper where yes you talk about the kind of Center uh I'm butchering this yeah Sur that's like the early 80s I think yeah yeah so that was one of the first applications of predictive coding was instead of Prior information possibly at a very deep level right from like whatever evolutionary priors or or cognitive priors it's just like well we sort of embody the prior that um this cell only needs to fire if what it's doing is is different from what would be

expected given its neighbors yes um yes so yeah and and so so and certainly and and one of I think Carl's earliest paper on this or one of the early ones I'm familiar with it's it's called something like a theory of qu responses responses thank you um also I've memorized them all not that I've read them but I've memorized them that's what you gotta do I I need to um up my game on that no so so yeah he discusses the role of lateral connections um and sort it's it's you know it's it's sort of a mechanism for competition in a way it's it's another way of inducing sparsity right so if you I've around with um versions of a hel Holtz Mach machine I don't know what you call this but I just added some lateral connections to the helmoltz machine because it doesn't have those right I mean well you break certain theoretical guarantees if you do that for but but anyway um and what happens is right you get a increasingly sparse representation because of this sort of competitive um potentially Behavior among the units at one level yeah yeah that makes sense I guess um I feel like an issue that I've always had with predictive coding at a very very general level and this is more definitely more indicative of my intellectual limitations than the actual theories limitations is it's very difficult to uh be able to say at a snapshot in time what's going on I think over a dionic dionic it kind of makes sense right like I get these bottom you know you get these bottom up prediction errors which in some ways lead to uh model changes at very you know much higher levels but still those higher levels constrain the expectations at the lower levels but I guess it's kind of like we're it's it's difficult to picture it as happening all at the same time because it's prediction error trickling up but the predictions that are going to deal with those prediction errors are already coming down so why are those prediction errors coming up in the first place right if they should be getting suppressed by the things that are coming down and maybe some people skirt this problem just by adopting some kind of linear time sequence in their minds I understand it's all happening at the same time but they go oh like this this level can't deal with it so it goes up and eventually you get the prediction and it suppresses it I've always just found that to be very problematic because who is saying at which level the prediction error stops right it just seems like it's a kind of Miracle so maybe you can help me or maybe you can't help me uh is this an intellectual is this an ual limitation of my own or is this something that's come to mind before um yeah so I think I can help you to some extent um I don't know that it's an intellectual limitation I think it's I don't see any problem I sort of I tend to think about this stuff sometimes in a quite visual way I just sort of Imagine imagine the stuff working and then also they try it out and code and it works so it's okay yeah um so I would like to give you a more discursive answer though I think um um so yes indeed if you try to so and I've I've definitely built models

um there's David Heeger has a paper also on it's a similar title to Carl's paper it's a theory of cortical something uh where he has a very similar model and yeah so so in implementing this it's like well what do you update first first of all and like how do you write so I think the solution to all that stuff part of it is and I thought a bit about how to how to build predictive coding architectures is um just do the updating asynchronously which you know if if you want a little bit of um background of that it's basically like each neuron each model neuron would just be its own sort of process like so we can go deeper into the the computer science stuff if you want but like basically each neuron would be running in its own sort of processor thread and it would take inputs and operate them as it gets them and then produce outputs as it as it has them to produce and all of this would be going on simultaneously so the idea that you need to first get all of the input to a to a layer the prediction error and integrate it with the top down predictions and figure out how much information to pass on in practice these are just not issues right it's so and even without going into the implementation of this stuff in terms of parallel distributed processing you can just think about um the energy function so the energy the free energy function that you're trying to functional that you're trying to minimize um it's it's additive across the nodes so each node computes its free energy so whenever it gets an input whatever its current state happens to be um right you just you just modulate its output based on that um so I guess and I guess I imagine this happening so it's it's sort of chaotic I suppose right it's um you would just basically each each each neuron could locally process its own its own input and produce an output and and then you know if you're simulating this on a Serial system you have to choose when to do what right um but that's not terribly Central I think to the architecture yes I guess I guess the problem there is if every neuron's just doing its own thing and over time you know over time over the sequence of neuronal activity prediction error just gets sliced off right and so you get this additive minimization of free energy you can run a kind of philosophical thought experiment and just go well what happens if the total you know amalgamation or accumulation of all the neurons Rel to this activity just they don't do they just don't they just can't get rid of all that prediction error but with we're siphoning off learning right so we're siphoning off uh parametric updating because we're saying that's happened asynchronously I'm just left with a bundle of prediction error and you never hear in the fre you know you always say well that's just going to end up get minimized so but yeah so I mean so it's your prediction error is never minimized right if were minimized minimized in the sense of like life would be over you would be dead yeah um so you know there's always it's an inter question you bring up like what what if there's just prediction error that can't be

handled is it's just going to be this oscillating like you know chaotic system where the prediction error just bounces around yes like that's what would happen like right you just you just shunt around the prediction error and you'd have sort of um and I've seen this happen in models right where like you just it just doesn't converge right you just sort of you just you just can't um there's no it's not a stable State it's maybe a maybe a um metastable state or something right where where um there there's there's oscillatory Behavior something um which Carl would say is is important right on to to life on larger time scales yes the Sol flows yeah so everything is always sort of smooth so it's it's it's pathological if you ever get a probability of a one or zero um these I'd say in real life as well exceptions like there might be some we can get this go straight into like um crazy amounts of mysticism maybe there's some kind of fire of one of like existence or something but like when it so I think I think so actually but yes in terms of the probabilities that are represented by your model like it's it's it's never going to be no prediction error um think that's a valid point and I guess once you fold in the the learning whether that's parametric learning or structure learning you know that prediction era that was so grave at one point is eventually going to get resolved one way or another whether that's through um you know I find sometimes when I run into this problem just anecdotally I find myself going towards a variational free energy for uh equation just being like there's my you know there's my safety because through either action or perception we're minimizing you know we're going down those uh descents of of free energy okay you you've you've s of you've quashed my anxiety somewhat um one other sticking point I have with active inference and uh this really came to light actually a couple of weeks ago when I spoke to Ronald Sadki who's a neuroscientist at University of Vienna and he's written he's written it's not a word he's written about the amygdala from an active inference perspective and he does this really nice thing of distinguishing between kind of I think it calls them preference priors you can call them phenotypic priors prior what you call them but prior preferences and then also what I've been calling State priors right so literally what state do I expect to be in now right and you can have that as your D Matrix or your B Matrix if we're doing pomdp schema and then also and and that obviously feeds into perceptual updating just on like some if we have just like naive Bays I know it's not happening like that but let's just say but then you also have these preference PS right in many ways as Cole has made very explicit like that too is shaping your perception right hence why we have visual Illusions I just find in the literature there hasn't mean too much work gone into distinguishing those and also talking about that interplay for you does that just come out in a hierarchical system that do those

things Converge on being the same thing we just call them a prior or is it useful to continue to distinguish them yeah that's a great question I mean ideally for me it's the um I guess the former it's it's that come out to be the same thing um so if you're doing pomdp modeling of course yes you'll Define like a c Vector which is your preference and then you you might also have um well you have you'll also have a d a d vector and so on but I think ideally in a hierarchical system it should be that the these are just you just have your priors and to me what the what the phenotypic um distribution is or the the the Deep sort of priors that represent desires and you know ostatic um requirements to me that's just when your your whole hierarchy links into this something that's sort of at the root no this of this graph right so you have like a basic need for food let's say that's not going to change right you can't really rewrite that or learn learn your way out of it um and then what's go what you're modeling at the lower levels could very well be a consist a situation that's not consistent with that prior being satisfied um but they're still just priors right and really what you want is that the priors at that deep level which are constitutive of and defining of the kind of thing you are you want those to smoothly propagate to the periphery for everything to work out well but of course what you're currently representing at the lower levels of of a hierarchy internal hierarchy is going to be the result of variational inference right so I think of I mean there's some talk about this in the later sections of that paper that Maxwell and Ryan Smith and I wrote on beliefs and desires um where basically the Q distribution is your sort of your beliefs in in the barebones version of the picture um yeah you're that's that's your estimate of the current state of the world it might be arbitrarily different in principle from your desires um but still um yeah and and the prior the priors are just the the generative model itself uh The Joint distribution that defines this kind of thing that you are um so I think that said a lot but I think basically the uh the distinction between two parameters C and and um D in a hierarchical model which is a way that you might set some of these these priors I think that that's more of a modeling convenience than anything I'm not sure that everyone would agree with me on that but it's it's really interesting uh and it's a really useful explanation I mean yeah once you have them in a kind of smooth hierarchy you you can start to kind of visually or mathematically see how it might all sync up I guess the issue is in folk psychology we we still there there's an intuition always that beliefs preferences let's say uh sorry desires preferences are distinct from my beliefs about what's going on in the world right now and I guess that's held in some sense because there's a hierarchy it it is yeah sorry I wouldn't say that this eradicates the distinction I I understand the the the pull towards that and this and like some of Andy Clark's uh where he's

um I don't I think he's he's it's not clear that he wants to preserve this this sort of classical distinction between beliefs and right Desires in a way um so I understand the pull towards that but I do think even though your your desires are these priors which are belief likee in a way because they to me they're they're really not right so the the prior the priors are right okay so the priors are certainly they have they they have they represent something something right and so they have you know if their desires then in the classical philosophy of Mind way of putting it they have satisfaction conditions right which are a lot like truth conditions it's just that it's a different propositional attitude so um you know they're they're in the same the representation wise they're in the same format it's a different attitude to the same kind of content yes and then the result of posterior inference is a belief and so the same structure cognitive or otherwise I tend I tend to think that there's a very transparent relationship between cognitive and physical sort of implementation structures anyway the structure can as considered as post result of posterior inference it's a belief um but then considered in terms of its Downstream effects in in a top down generative model it sort of functions as a desire so you have things in the system yeah in yeah I mean I wonder whe How yes the hierarchy is a is a really useful way in which that I guess again philosophically vague distinction can be made concrete and obviously people also talk about the temporality aspect of the generative hierarchy in so far as the higher levels track slower fluctuations in the external Dynamics and the lower levels the faster Dynamics it makes one think whether that that necessarily maps onto desires preferences and so on only being at the higher level that IE being the result of slower fluctuations right like I had ice cream I've had ice cream a thousand times I like ice cream versus a belief that there is ice cream or that there isn't ice cream it's an interesting question it's an interesting question whether beliefs sorry well desires let's say necessarily need to originate at the higher level and whether sort of more classical folk psychology beliefs a lower level oh can I jump in on that no no no I wasn't just smitb please yeah yeah yeah go I really like so I think I think Andy Clark also talks about this in one of his recent papers more recent papers so there's a sense in which you really can talk about sort of like false desire and versus more genuine desire that really stems from your being in that I think sort of genuine sorts of desires are from the deepest levels of the model but then in cases of like addiction what gets happens is yes right so so basically there's a bottomup sort of an attractor sort of manages to establish itself in your system on the basis of bottomup input that creates a a sharp expectation over a particular outcome that might be not at all a good fit with your deeper priors but that just gets self-confirming account of um of addiction and there's a desire there that will in that it motivates

your behavior and you will go for the thing but uh it's definitely deeply obviously um it it impoverishes the rest of the model and right oh that's really nice yeah that's really really nice um John Vu is a very you know big hero of mine talks about reciprocal narrowing and reciprocal opening with respect to addiction I think this isn't his direct work I can't remember whose it is but the idea being that as you said addiction is a kind of downward spiral such that and you can just think about in terms of dopamine um such that a singular Source becomes your kind of source of pleasure in the very folk psychology way but also like your singular prior and everything becomes determined by that and maybe well he he you know he will use the word like Agape but you know you have this kind of opening maybe Hy call Alfa right you have this kind of disclosure and that actually opens yourself up to the world and I guess yeah maybe is a bit mystical but I wonder whether it in doing that we're uncovering something which is actually was already there in a way I mean absolutely I think I think the sort of the sure there's a there's a mystical take on that but it's it's also just um absolutely the case that you want you want as much variability in your States as as is consistent with preserving the phenotype and so y that's something that for example we've written that uh ma and Maxwell and and um um ready and others have written in that resilience uh the work on resilience Mark Miller and yeah yeah Mark yes thank you um I've been doing a lot of things so I haven't thought about that paper in a while but uh but yes like and that just comes out of just VAR I think just variational inference even like you just you just want to maximize the the entropy of your distribution subject to the constraints from your model and yeah Ryan who you mentioned earlier he sort of has this idea of um structure learning aiding emotional granularity and that being really useful because you don't get trapped into thinking that you're Furious if you're maybe just a bit frustrated or and and and so on it's all uh complexity minus accuracy at the end of the day um yeah I don't know what I've had this idea and I'm think you may be writing a paper on it but I there's I think a really nice addition to this whole picture is the idea of selective attention um so Beck mitza who I think was in Carl's lab back in the day he wrote a bunch of papers with Carl about attention and he has this 2019 paper about selective attention I've spoken about on the podcast before and the idea being that your priors your preferences constrain your attention and that kind of make that's very sort of in sync with traditional um biased competition Theory right and is actually stuff that I wrote about during my masters the idea being that you know what is elevated in terms of gain control aligns with is actually in line with your goals so if I'm looking for a ginger person then all the templates of things that are orange or Ginger stand out to me and I've been having this idea that we can

actually root that selective attention to the very deepest level of our belief hierarchy such that we kind of do have it's very Jordan Peterson actually because it's very sort of narrativity but we do have these kinds of deep like what like we have basically a guiding principle and that could be governed by your biology by your so by your soou culture but I really like this idea that like your value you know I tweeted that thing that you liked which was Simone Weil attention is prayer um influenced by mcil Chris what attention is a moral act and I think that's really powerful because attention is basically a phenomenological indicator of that which you are Guided by so I'm thinking maybe again it's a I'm putting this out there so no one steals it but I've got this I'm trying to think about how to formalize this idea that like the very deep temporary invariant parts of our being might actually be the guiding force of our attention scha that's that's wonderful I hope you write that paper um I I haven't done so much research on attention as such uh in in psychological literature and such so it's it's it's really cool to hear that what you've researched confirms this kind of perspective well hopefully hopefully someone doesn't come into the comment section you know are those comment sections but people always have things to say um this might be a slightly selfish question uh but I don't think it is because I think people also benefit from this you say that I wanted to kind of go back to your Journey as an academic if you will and starting off as philosophy and then Andy Clark and then sort of finding modeling saying what really helped us actually modeling a Helm Holtz machine for example and you hear this a lot you know if you want to learn how to do pomdp stuff know how to do the mat lab work the question I've always had like I haven't done any of this coding what does it look like I mean when you say I code a Helm Holtz machine what do you get on your screen I know that sounds really stupid but I think people are kind of confused about what that looks like especially when K talks about oh you get these little agents who are doing things what what does that actually look like yeah no I don't it's not a stupid question it's it's kind of it's one of the problems you have to solve if you wanna want to do this stuff right so so what I did I mean this was in the relatively early days of this stuff in academ in broader Academia in the sense that unless you were doing machine learning you probably didn't usually care about this stuff so I did was I started writing it in JavaScript I used Dynamic CSS to render this as a web page you know um I'll have to send I'll have to send you some link I'll post it on my Twitter or something yeah basically I had this very early the reason I never shared it so much with the world at the time is that it didn't work very well at all but I basically had a Helo machine visualized in a in a browser and it was like you just have like um some squares for each and on there would be some in the hidden layers you'd have these sort of

staticky looking patterns as you'd expect and then you'd visualize the output layer as an image and I sort of overlaid the um the input the ground truth input image with the prediction using like red and blue or something I got those color codes come from like Jeff hinton's work and a lot of people have used them um or red and green maybe but anyway um so that to me that I really needed this concrete visualization to like understand what's going on if you run Carl's mat lab code it's much more like you just get the output I mean it's not it's a that's a fine way of working it's worked well for for many people right like you see the belief distribution change over time um just sort of plotted on um on an axis like you know uh based on various conditions um but so yeah so in in any case I mean you I think you you need to set up um a visualization that makes this stuff intelligible I feel like I don't know if that was the hard of your question but you no no no no no I'm yes I'm just kind of I guess people always ask like what what are these like what are you know when we talk about insilico agents right is that a thing like really or is that just a metaphor I mean I kind of hope it's a metaphor because I just you know kill thousands of them in a year if they're real you know oh yeah yeah yeah yeah yeah sure my browser window and it's just gone um but no I mean I think to the extent that you can model to the extent that you think these types of models are models of agency yeah it's real um in your browser window what what do you actually get you know what I mean like what what do you see when you say agent I think people actually think of little Gummy Bears or you know little teddy bear looking things okay well so the helmos machine then is a bad example because that's that's really just doing sort of uh it was really just showing online learning of like um of a generative model and it doesn't an action component to be interesting but um I've seen tons of visualizations of um of active inference models and other similar models so I mean if you look at some of the um some of the machine learning work on learning to play video games is a good example so so that's a strange case but you know if you have a a model that's trained to play Super Mario Brothers then I mean you see the effects of the agent anyway um you can you can just visual it depends on your problem right there's plenty of um sort of grid World type simple proof of concept type active inference models that get made where you can see the location of an agent on a little map or something um I love that yeah but but typically honestly I mean just as a as a matter of like everyday sort of um working reality it's like usually you'll get a series of plots of the belief belief updates over time and and the action that was chosen right and that works yeah and it's an ongoing struggle to find better more powerful visualizations for this stuff sure sure I imagine it is okay sort of just getting you know wrapping things up you've gone from philosophy to computational science

let's say um and the the kind of contentious Bridge there let's say between phenomenal structures and mechanical explanations or mechanical formalisms is often called de generative Passage um started I guess by verel and then formal or you know that term instituted by Maxwell and his team in that 2022 paper I really like that you I think I think I get this from a lot of philosophers who start off as philosophers is the kind of non- reducibility of intentionality the non- reducibility of Consciousness where what do you think of the generative passage and what do you think is the sort of most I won't say likely but the most adaptive or effective manifestation of it and by that I mean some people talk about ontological naturalism which is we could basically posit as a reductive approach an epistemological naturalism which is that the way that we come the way we think about or describe um let's say in this case basian Dynamics is going to be helpful um or actually oh no not in that case it's not helpful it actually does describe the structures the intentional structures or procedural or more you know mechanical way where it's actually just useful where it's kind of instrumental um but actually might not be telling us anything too concrete about the phenomenal reality now since you've kind of flirted in both camps where uh where do your allegiances lie I don't mean what do you think yeah no uh so as far as the narrowly the term generative passage um I stop me if you think I'm misusing that term uh I haven't been deeply involved in thinking in that term in those terms but I think you're talking about the relationship between phenomenology and uh in this case basian Dynamics let's say or or um well I mean people talk about neurophenomenology right so it could be neuronal Dynamics yeah yeah great so um yeah so my overall so there's a lot of things rolled into that question um epistemologically I kind of see things as and this it's good that you brought this up I wanted to comment on this earlier um so another paper that I haven't published is on is on this topic so um like I kind of I would be a transcendental idealist except that the except that the evidence that I've seen for that position is empirical so I kind of I probably someone said something similar but I want to I want to write a paper on that a sort of cont like almost empirically informed conis where it's like well what we've learned about how these systems work um provides evidence for the conclusion that the our experience of reality is is mostly based on prior structures in the brain right um so I think that's really interesting epistemological it's different from sort of classical transcendental idealism where you just had K and categories right um because you're you're sort of presuming to say something it's there's a sense in which the the the naturalistically studi reality comes first there because you're saying it's like well the way neurons work we probably en code these generative models in our brain right and that's why we conclude that um we

have subjective ontologies and such that we learn in order to parse out the sensory data um so so overall that's my that's my sort of view on the epistemology of it but um in terms of the relationship between phenomenology and cognitive structures and physical structures um I Tred try to in my work I try to um argue for the the the most minimal account of that relationship where where transparency is the ideal um across all those levels so of course that's not going to work perfectly like if I see a dog on the street doesn't mean there's a dog in my head it doesn't mean that something um it does mean that there's something whose Dynamics reflect the Dynamics of that thing out there um if there's a thing out there as we said earlier um but but let's for a second let's just think about the relationship between the phenomenology and the cognitive structures right um um so yeah also I cite some work by the gestal psychologists who really assumed that there was a strong isomorphism there um that you really in a sense if you if you looked closely enough at the um well I sorry I was I was shifting to the physical level of description again but certainly that there you should be able to sort of read understand the structure of phenomenology in terms of um in terms of the structure of these basian prior prior structures basically and that get updated by by sensory experience and so I totally agree with Maxwell at all on that the one reason I haven't been more gung-ho about um computational phenomenology is just that I already assume something further which is that like so so their their research program as I understood is let sort of step back and just try to build a generative model for phenomenology yeah and understand how you understand phenomenology in terms of general of modeling I think you can just shoot straight through the cognitive structure and just talk about isomorphism between the phenomenological structures and the cognitive structures and the um implementation bases for those inter so so I I I guess it's good it's epistemically humble and it's a good project to try to do that um I just find that it's more effective for me to be constrained by more um the so I try I I usually when I think about this stuff I'm constrained by thinking about neuronal implementation level as well as cognitive level as well as phenomenological I was hoping you gon to be looser oh well I was hoping you got it's there might be some link um okay I'm joking you're entitled to of course you're entitled I mean I I'm personally I mean I've written about this a little bit publicly uh ontologically speaking I'm quite agnostic um I have a I have maybe a slightly romantic um constraint which means that I just don't want to do the reductionism I just don't um but actually I should clarify I've actually got the paper up in front of me uh Maxwell and colleagues make it clear that actually the so-called ontological naturalization doesn't necessarily entail reductionism um only so only that and I can quote here some theal framework means that one reformulates all

the properties entities and processes postulated by that theory in terms of those posited in the ontology of the empirical Sciences um and how that gets then fleshed out in terms of epistemological naturalization is that you would take the core principles and forms of your explanation from the Natural Sciences and then you would just have methodological naturalization what I call procedural naturalization before which is bringing your sort of methodology into continuity with the empirical Sciences um so as you say kind of constraining phenomenology uh you know according to the principles of Psychology and Neuroscience without saying that they are those principles or without saying that we should only understand them by those principles um yeah it's a really it's a really nice trifect to that um and I guess there isn't an answer but you know you have some people who are very adamant that perhaps even the methodological naturalization is a step too far um and we should just be as humble as possible with respect to the nature of conscious and say you we just have no idea in what form it's grounded or in what form it relates to the physical universe but I I I'm probably not that loose but it's inter can I follow up with one more please um because I I think it's a really important Point um so I I I don't think in reductive terms about Consciousness like so even if even if it turns out that there's a perfect isomorphism between phenomenal cognitive and physical structures uh I don't think that means that everything would be reduced to physical structures like so so I see more I hope that these lines of inquiry would converge so that you'd end up seeing that these are different ways of talking about the same thing but it doesn't mean that one of those ways is fundamental right um so yeah I don't know if that is any different than what you just said but yeah yeah yeah yeah no that's absolutely great um okay final question uh actually penultimate question we can rush through these uh pan psychism is it is it is it or is it brilliant sorry you put it that only because you mentioned it only because you mentioned it earlier no no it's certainly not um I I don't think it's I don't think it's brilliant to just rush headlong into it without thinking about it but um no I think I think it depends what you mean right but you assume we're talking about like phenomenal Consciousness I'm quite comfortable with the idea that any physical structure um there's some edge cases like rocks where it's like well you it's kind of like a don't care case as Quin would say it's like yeah um it's Ed where there's no there's no structure to speak of I guess I guess to me the natural view is that the kinds of contents you can represent consciously depend on the complexity of your structure but the base ground existence of sort of a field in which these things are represented I don't see why that should be different from like the basis for physical existence good final question yeah you can only have one for the rest I'm being

trivial now but you can only have one for the rest of your life philosophy computer science man which one are you keeping oh that's rough um I was going to say computers science CU I think that the best computer science contains good philosophy but I think in terms of the intention rather than the extension of these things I think philosophy because you'd end up inventing computer science if you did philosophy well enough in the long term fantastic I'm so glad you said that uh which which one's harder um computer science is way harder is it I mean sorry if hard means like it's hard to do it's like a struggle then yes but if you mean like cognitively theoretically oh then philosophy good good yeah yeah we'll have to discuss more about uh about all of this stuff this was so much fun Alex thank you uh thank you for dealing with my Layman thoughts and terrible attempts to trying to understand things that are way Bey on my pay grade but this was awesome likewise thanks Darius this was this was great cool all right thank you everyone for tuning in as well okay